

Problem 2

a) A possible padding technique would be adding for example an "end message" marker, let's say Q and then "x"'s until we fill the 9 positions block.

Ex) FATHER Q x x
ICE Q x x x x x

If the block is complete (has 9 letters) we send another block with only Q x x x x x x x x x to mark that and be consistent with the padding.

Ex) ALAMBICAT Q x x x x x x x x

At decryption, just remove all 'x's from the end until you reach the Q marker. Remove the Q marker also.

b) ALAMBICAT UL
1 block 1 block

~~We have to read~~
ALAMBICAT
step 4 →

A	M	C
L	B	A
A	I	T

 step 5 →

A	M	C
B	A	L
T	A	I

 step 6 →

A	B	T
M	A	A
C	L	i

 step 7 →

T	B	A
A	A	M
i	L	C

 → T B A A A M i L C
cryptogram.

ULQxxxxxx
step 4 →

U	x	x
L	x	x
Q	x	x

 step 5 →

U	x	x
x	x	L
x	Q	x

 step 6 →

U	x	x
x	x	Q
x	L	x

 step 7 →

x	x	U
Q	x	x
x	x	x

 → x x U Q x x x L x
cryptogram.

c) ALAMBICAT padding
1 block 1 block

ALAMBICAT → already done → T B A A A M i L C

Qxxxxxx

Q	x	x
x	x	x
x	x	x

 step 5 →

Q	x	x
x	x	x
x	x	x

 step 6 →

Q	x	x
x	x	x
x	x	x

 step 7 →

x	x	Q
x	x	x
x	x	x

 → x x Q x x x x x x

(Problem 2 - continued)

d) AES → Rijndael algorithm → shift rows operation

e)

$\xrightarrow{\text{step 4}}$

1	4	7
2	5	8
3	6	9

 $\xrightarrow{\text{step 5}}$

1	4	7
5	8	7
9	3	6

 $\xrightarrow{\text{step 6}}$

1	5	9
4	8	3
7	2	6

 $\xrightarrow{\text{step 7}}$

9	5	1
3	8	4
6	2	7

→ 9 5 1 3 8 4 6 2 7

Direct permutation
 ⇒ ~~from table~~:

$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 9 & 5 & 1 & 3 & 8 & 4 & 6 & 2 & 7 \end{pmatrix}$

f) Inverse permutation

$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 3 & 8 & 4 & 6 & 2 & 7 & 9 & 5 & 1 \end{pmatrix}$

→ This is obtained from the direct permutation.

Algorithm:

- 1) Take each position from inverse: let's say 1
- 2) Look in direct perm. on line 2 and search for
- 3) Look on the same column in dir. perm on ~~line~~ first line and take that value